

Assignment 6: Graph Neural Networks [Implementation]

Spatial Graph Convolutional Networks for Traffic Speed Prediction

This project uses a Graph Convolutional Network (GCN) to predict traffic speed using spatial sensor data. The dataset used is METR-LA, which contains traffic speed data collected from road sensors in Los Angeles. Each traffic sensor is represented as a node, and spatial connections between nearby sensors are represented as edges.

The goal is to predict the future speed of each sensor using:

- The sensor's historical traffic speed
- The spatial location of the sensor
- The traffic behavior of nearby sensors

Questions

1. Does spatial graph information improve traffic prediction? Compare a GCN with a non-graph model such as an MLP.
2. Does increasing GCN depth always improve performance? Compare GCN models with different numbers of layers. This helps analyze whether deeper GCNs cause over-smoothing.
3. Are distance-based edge weights useful? Compare weighted and unweighted graphs. In the weighted graph, closer sensors have stronger edge weights.

Experiments

1. Experiment 1: Spatial GCN vs Non-Graph Models

Compare Historical Average, MLP and Spatial GCN. The goal is to determine whether spatial graph structure improves traffic speed prediction.

2. Experiment 2: Effect of GCN Depth

Compare GCN models with different depths:

Required plots:

- MAE vs number of GCN layers
 - RMSE vs number of GCN layers
 - Average cosine similarity vs number of GCN layers
3. Experiment 3: Weighted vs Unweighted Graph

Compare Unweighted graph and Distance-based weighted graph.

4. Experiment 4: Spatial Error Analysis

For each sensor, compute its average prediction error. Analyze:

- Which sensors have the highest error?
- Are high-error sensors located in specific regions?
- Do central sensors behave differently from peripheral sensors?
- Are isolated sensors harder to predict?
- Does the GCN help more in dense sensor regions?